Stack and architecture

Browsers → Render HTTPS → Express API → Turso hosted SQLite. React 19, TypeScript and Vite provide order entry, queue, guest QR tracking, ready board and lead-only menu/reports. Tailwind CSS 4 and shadcn/ui supply the UI; Zod validates API inputs. Node.js 24+ runs Express 5 with the libSQL HTTP client. Local development supports built-in SQLite with WAL. Transactions assign increasing tickets, validate prices and transitions, and preserve change history. Reports use Singapore dates and accepted prices; payments stay on the organiser's POS.

Offline operation and sync

Staff sign in and load the built app/menu online first. A service worker caches the app shell; Dexie/IndexedDB holds snapshots and a durable outbox. Offline resume requires an unexpired role hint. Saved orders/status changes survive reloads. Existing tickets advance locally; new disconnected orders receive their number and guest QR after server confirmation.

With the staff app open, sync runs after actions, reconnection, every 15 seconds or manually. Commands replay sequentially; UUID retry keys prevent duplicate orders, and state/revision checks reject stale changes. Server acknowledgements are saved before commands are removed. Network/session failures retain work; rejected changes remain visible for review. Guest pages and the board poll every three seconds and see changes after sync.

Personal data and PDPA handling

The app stores first names (up to 40 characters), drink details/prices, ticket IDs, status, timestamps and private guest tokens. Change history includes previous orders, reasons and staff roles. Copies exist in staff IndexedDB, CSV exports and backups. No phone numbers, email, NRIC or payment credentials are collected.

The guest privacy notice explains fulfilment, public display, copies and booth-lead enquiries. The board shows ready names/numbers; anyone holding a random 192-bit guest link can view its order. Staff APIs require signed sessions, with lead permissions for menu, reports and corrections. Password checks use scrypt; cookies are HttpOnly, SameSite=Strict and Secure in production. Sessions and local role hints expire after eight hours; API responses are excluded from service-worker caching.

Manual, preview-first cleanup anonymizes closed orders before an approved Singapore-date cutoff, revokes links and removes identifying history while retaining reports. It does not erase provider copies, device caches, CSVs or old backups. Sign-out leaves IndexedDB intact. There is no automatic retention or app-level encryption at rest. Organiser approval of notices, consent, DPO contact, retention, data-request/breach procedures and overseas hosting remains pending; PDPA compliance is unassessed.

Known limitations

Fresh login, lead edits, Undo, report refresh and CSV exports need connectivity; disconnected devices cannot share updates. Unsaved drafts can be lost on reload; browser-data clearing/eviction can lose pending work. Guest sound requires an open, awake page; background push and Telegram/WhatsApp are deferred. Shared role passwords do not identify individuals. Reports show recorded value, with manual POS reconciliation. Render Free has cold starts. Public end-to-end tests, hosted recovery, capacity and physical-device QR/audio/network rehearsal remain pending.